

Radial and Angular Risk in Stochastic DC Optimal Power Flow:

How Distributional Misspecification Relocates,
Rather Than Merely Inflates, Estimated Grid Risk

Pascual Zárates*

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Abstract

We study a Monte Carlo adequacy experiment in which a fixed transmission network is solved under $N = 10,000$ independent realisations of nodal demand. The dispatch model is a DC optimal power flow augmented with load shedding priced at the value of lost load, which renders the program feasible for every demand realisation and converts “infeasibility” from a Boolean solver outcome into a measurable quantity in megawatts. We establish four structural results. First, the recourse formulation has relatively complete recourse, and shedding is optimal only when the zero-shed program is infeasible, provided the value of lost load exceeds the largest nodal dual—a condition strictly stronger than exceeding the costliest generator, because congestion rents can drive locational prices above every unit’s marginal cost. Second, under independent exponential sampling $D_i = \mu_i E_i$ the *system-level* coefficient of variation is $\sqrt{\sum_i \mu_i^2} / \sum_i \mu_i$, which lies in $[n^{-1/2}, 1]$ and is governed entirely by load concentration. Third, the servable set along any demand ray is an interval $[0, T^*(s)]$, so adequacy failure is exactly the event $\|D\|_1 > T^*(D/\|D\|_1)$: adequacy is an *angular* quantity. Fourth—and contrary to our initial conjecture—line congestion is very nearly *radial*: on a 9-bus model of the O’ahu system a single magnitude threshold $t_c = 585$ MW reproduces the congestion frequency under two radically different demand distributions to within 0.6 %. We then show empirically that replacing the assumed $\text{Exp}(1)$ marginals with 8760 hours of filed utility load data drives estimated shedding probability from 19.77 % to exactly zero while *raising* estimated congestion frequency from 58.40 % to 92.51 %. Misspecification therefore does not simply inflate risk; it moves risk from one failure mode to another. Finally we note that nameplate capacity overstates deliverable capacity by 28.3 % on this network, reducing the apparent 43 % reserve margin against the observed annual peak to 2.63 %.

Keywords: DC optimal power flow; stochastic programming; relatively complete recourse; resource adequacy; model misspecification; Monte Carlo simulation.

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1 Introduction

A deterministic optimal power flow answers one question: given this network and this demand, what is the cheapest feasible dispatch? A great deal of practical interest, however, attaches to a different question—given this network and a *distribution* over demand, how often does something go wrong, and what exactly goes wrong. Answering the second question requires re-solving the first many times, and the answer one obtains is a functional of the assumed demand distribution as much as of the network.

This paper examines that dependence in a setting where the “true” distribution is available. We fix two networks—the IEEE 14-bus test system and a nine-bus model of the O‘ahu (Hawai‘i) transmission system assembled from utility filings—and solve each under $N = 10,000$ demand realisations. We ask the three questions a planner would ask:

(Q1) How often is the most expensive generator dispatched?

(Q2) How often is the problem infeasible, i.e. how often must load be shed?

(Q3) How often do network elements reach their thermal limits?

The demand model specified at the outset is the natural first choice in the absence of data: at each load node i , draw $D_i = \mu_i E_i$ with $E_i \sim \text{Exp}(1)$ independent across nodes, so that $\mathbb{E}[D_i] = \mu_i$. For the O‘ahu system, however, we possess 8760 hours of filed hourly load. This permits a direct comparison between the answers the assumed distribution produces and the answers the empirical distribution produces, on a network that never changes.

Contributions. Our contributions are of two kinds. On the structural side (Sections 2 to 4) we characterise the feasible set of the recourse program and the geometry of its two failure modes. The central structural observation is a decomposition of failure into a radial component (the magnitude $\|D\|_1$) and an angular component (the direction $D/\|D\|_1$), and the finding that the two failure modes load onto these components very differently. On the empirical side (Sections 7 to 9) we quantify the misspecification effect and show that its sign is not constant across failure modes.

A note on how these results were obtained. Proposition 4.5 contradicts the conjecture with which this investigation began. We had expected congestion to be the directional quantity, on the intuitive grounds that a line binds when power must cross it, and whether power must cross it depends on *where* demand sits. The conditional experiment of Section 4 refutes this cleanly, and the corrected statement—that adequacy is angular and congestion is radial—is both simpler and better supported. We record the failed conjecture because the diagnostic that settled it (conditioning on total demand and comparing failure rates across demand models) is reusable.

2 The dispatch model with recourse

2.1 Network primitives

Definition 2.1 (Network). A *network* is a tuple $\mathcal{N} = (\mathcal{B}, \mathcal{L}, \mathcal{G}, b, f, t, \bar{F}, p^{\max}, c, g, r, \beta)$ where $\mathcal{B} = \{1, \dots, n_b\}$ indexes buses, $\mathcal{L} = \{1, \dots, n_\ell\}$ branches, $\mathcal{G} = \{1, \dots, n_g\}$ generators; $f, t : \mathcal{L} \rightarrow \mathcal{B}$ give each branch its endpoints; $b_\ell > 0$ is branch susceptance in per unit; $\bar{F}_\ell > 0$ its thermal rating in MW; $p_g^{\max} > 0$ generator capacity; $c_g \geq 0$ generator marginal cost in $\text{MW}^{-1} \text{h}$; $g : \mathcal{G} \rightarrow \mathcal{B}$ assigns generators to buses; $r \in \mathcal{B}$ is the reference (slack) bus; and $\beta > 0$ is the power base.

We work throughout with the standard linearised (“DC”) power-flow approximation: branch resistance and reactive power are neglected, voltage magnitudes are held at unity, and branch flow is proportional to the phase-angle difference across the branch.

Assumption 2.2 (Standing assumptions). (i) The network is connected. (ii) Generator minimum stable output is set to zero, $p_g^{\min} = 0$ for all g . (iii) Generator cost is affine with zero fixed term, so total cost is $\sum_g c_g P_g$.

Assumption 2.2(ii) deserves comment because it is consequential. The O‘ahu fleet described in Section 6 carries 531 MW of nameplate minimum stable output. Enforcing those bounds inside a continuous linear program, without the binary commitment variables that would permit a unit to be switched off, would render every sufficiently *low* demand realisation infeasible—for reasons of unit commitment rather than of adequacy. Since this study concerns whether the system can *meet* demand, we permit the fleet to shut down freely. Assumption 2.2(iii) discards the quadratic term present in the IEEE-14 cost data; we return to its consequences in Remark 6.2.

2.2 The recourse program

Definition 2.3 (Recourse dispatch program). Given a network $\mathcal{N}$, a demand vector $D \in \mathbb{R}_{\geq 0}^{n_b}$, and a penalty $\text{VOLL} > 0$, define

$$\mathcal{P}(D) : \min_{P, \theta, F, S} \sum_{g \in \mathcal{G}} c_g P_g + \text{VOLL} \sum_{i \in \mathcal{B}} S_i \quad (1a)$$

$$\text{s.t. } 0 \leq P_g \leq p_g^{\max}, \quad g \in \mathcal{G}, \quad (1b)$$

$$\theta_r = 0, \quad (1c)$$

$$F_\ell = \beta b_\ell (\theta_{f(\ell)} - \theta_{t(\ell)}), \quad \ell \in \mathcal{L}, \quad (1d)$$

$$\sum_{g: g(\ell)=i} P_g - (D_i - S_i) = \sum_{\ell: f(\ell)=i} F_\ell - \sum_{\ell: t(\ell)=i} F_\ell, \quad i \in \mathcal{B}, \quad (1e)$$

$$-\bar{F}_\ell \leq F_\ell \leq \bar{F}_\ell, \quad \ell \in \mathcal{L}, \quad (1f)$$

$$0 \leq S_i \leq D_i, \quad i \in \mathcal{B}. \quad (1g)$$

The variable S_i is *load shedding* at bus i : demand the system elects not to serve. We write $C(D)$ for the optimal value of $\mathcal{P}(D)$ and denote by $\mathcal{P}_0(D)$ the *zero-shed program* obtained by fixing $S \equiv 0$ (a pure feasibility problem after dropping the objective).

Definition 2.4 (Servable demand). D is *servable* if $\mathcal{P}_0(D)$ is feasible. Write $\mathcal{S} \subseteq \mathbb{R}_{\geq 0}^{n_b}$ for the set of servable demand vectors.

The role of (1g) is methodological rather than physical, and is worth stating plainly. A Monte Carlo study that omits shedding must detect adequacy failure by catching an **INFEASIBLE** status from the solver. That is a poor instrument on two counts: it is a single bit, carrying no information about the magnitude of the shortfall, and solvers return the same status for numerical difficulties unrelated to the physics. Including S makes the program feasible always, so that failure is reported as a real number—expected unserved energy, the quantity reliability studies actually use.

Proposition 2.5 (Relatively complete recourse). *For every $D \in \mathbb{R}_{\geq 0}^{n_b}$, $\mathcal{P}(D)$ is feasible and its optimal value is finite, with $0 \leq C(D) \leq \text{VOLL} \sum_i D_i$.*

Proof. Take $P = 0$, $\theta = 0$, $F = 0$, $S = D$. Constraints (1b), (1c), (1f) and (1g) hold trivially; (1d) holds since $\theta \equiv 0$; and (1e) reduces to $0 - (D_i - D_i) = 0$. Hence the feasible set is nonempty. The objective is bounded below by 0 since $c \geq 0$ and $\text{VOLL} > 0$, so a finite optimum exists by LP duality; evaluating the objective at the exhibited point gives the stated upper bound. $\square$

Proposition 2.5 is the property known in stochastic programming as *relatively complete recourse*: the second stage is feasible for every realisation of the random data. It is what licenses the estimator of Section 5 to be well-defined without conditioning on solver success.

2.3 Shedding as a last resort

Proposition 2.5 guarantees feasibility but not that shedding is used only when unavoidable. If VOLL were small, the optimiser would shed load opportunistically whenever generation happened to be expensive, and the reported “adequacy failures” would be economic choices rather than physical ones. The following gives a sufficient condition.

Proposition 2.6 (Shedding occurs only under infeasibility). *Let $D \in \mathcal{S}$, let $C_0(D)$ denote the minimum generation cost among zero-shed feasible points, and let $\lambda^\circ \in \mathbb{R}^{n_b}$ be an optimal dual vector for the balance constraints (1e) of that zero-shed problem. If*

$$\text{VOLL} > \max_{i \in \mathcal{B}} \lambda_i^\circ, \quad (2)$$

then every optimal solution of $\mathcal{P}(D)$ satisfies $S^* = 0$.

Proof. For $u \in \mathbb{R}_{\geq 0}^{n_b}$ let $V(u)$ be the minimum of $\sum_g c_g P_g$ subject to (1b)–(1f) with D replaced by u , and $V(u) = +\infty$ if no such point exists. V is the value function of a linear program in its right-hand side, hence convex on its domain, and λ° is a subgradient of V at D ; therefore

$$V(D - S) \geq V(D) - \langle \lambda^\circ, S \rangle \quad \text{for all admissible } S. \quad (3)$$

Let (P, θ, F, S) be feasible for $\mathcal{P}(D)$ with $S \neq 0$. Its generation cost is at least $V(D - S)$, so its objective satisfies

$$\sum_g c_g P_g + \text{VOLL} \sum_i S_i \geq V(D - S) + \text{VOLL} \sum_i S_i \stackrel{(3)}{\geq} C_0(D) + \sum_i (\text{VOLL} - \lambda_i^\circ) S_i > C_0(D),$$

the last inequality strict because $S \neq 0$, $S \geq 0$ and (2) holds. Since the zero-shed point attaining $C_0(D)$ is feasible for $\mathcal{P}(D)$ with objective exactly $C_0(D)$, no solution with $S \neq 0$ can be optimal. $\square$

Remark 2.7 (Why exceeding the costliest generator is not enough). It is tempting to impose the simpler condition $\text{VOLL} > \max_g c_g$. This is *insufficient*. The dual λ_i° is the locational marginal price at bus i , and in a congested network congestion rent can push it above every generator's marginal cost; in a companion multi-period study on the same data we observe locational prices reaching $381 \text{ MW}^{-1} \text{ h}$ against a costliest unit at $237.97 \text{ MW}^{-1} \text{ h}$. In the limit where a bus is electrically isolated by binding thermal limits, λ_i° is unbounded and no finite VOLL suffices. We therefore set $\text{VOLL} = 10,000 \text{ MW}^{-1} \text{ h}$ —roughly $42\times$ the costliest unit—and verify (2) *ex post*: across 6000 sampled realisations on the two networks, in every case where the recourse solution shed load, an independent feasibility test confirmed that $\mathcal{P}_0(D)$ was genuinely infeasible. There were no counterexamples.

3 Demand models

Throughout, $\mathcal{B}^+ = \{i : \mu_i > 0\}$ denotes the load buses, $n = |\mathcal{B}^+|$, and $M = \sum_i \mu_i$ the design mean total demand.

3.1 Independent exponential sampling

Definition 3.1 (Model A). $D_i = \mu_i E_i$ for $i \in \mathcal{B}^+$ and $D_i = 0$ otherwise, where $E_i \stackrel{\text{iid}}{\sim} \text{Exp}(1)$.

Each node then has $\mathbb{E}[D_i] = \mu_i$ and $\text{sd}(D_i) = \mu_i$, so the *nodal* coefficient of variation is exactly 1. The system-level dispersion is quite different, and is determined entirely by how concentrated the load is.

Proposition 3.2 (System coefficient of variation). *Under Model A, let $T = \sum_i D_i$. Then $\mathbb{E}[T] = M$, $\text{Var}(T) = \sum_i \mu_i^2$, and*

$$\text{CV}(T) = \frac{\sqrt{\sum_i \mu_i^2}}{\sum_i \mu_i}. \quad (4)$$

Moreover

$$\frac{1}{\sqrt{n}} \leq \text{CV}(T) \leq 1, \quad (5)$$

with the lower bound attained iff all μ_i are equal, and the upper bound attained iff exactly one μ_i is nonzero.

Table 1: System coefficient of variation under Model A: identity (4) against simulation (200,000 draws).

System	n	$n^{-1/2}$	Analytic (4)	Simulated	Upper bound
IEEE 14-bus	11	0.3015	0.4439	0.4446	1
O‘ahu 9-bus	8	0.3536	0.5337	0.5330	1

Proof. Linearity gives $\mathbb{E}[T] = \sum_i \mu_i \mathbb{E}[E_i] = M$. Independence and $\text{Var}(E_i) = 1$ give $\text{Var}(T) = \sum_i \mu_i^2 \text{Var}(E_i) = \sum_i \mu_i^2$, whence (4). For the bounds, write $\mu = (\mu_i)_{i \in \mathcal{B}^+}$ so that $\text{CV}(T) = \|\mu\|_2 / \|\mu\|_1$. The Cauchy–Schwarz inequality $\|\mu\|_1 = \langle \mu, \mathbf{1} \rangle \leq \sqrt{n} \|\mu\|_2$ gives $\text{CV}(T) \geq n^{-1/2}$, with equality iff $\mu \parallel \mathbf{1}$. The upper bound is $\|\mu\|_2 \leq \|\mu\|_1$, valid for nonnegative vectors, with equality iff μ has a single nonzero entry. $\square$

Proposition 3.2 is the reason a demand model with nodal $\text{CV} = 1$ does not produce a system with $\text{CV} = 1$: independent nodes diversify. How much they diversify is a property of the network’s load geography, not of the distribution. The two systems studied here sit at different points of (5), and Table 1 reports both the analytic value and a 200,000-draw simulation.

O‘ahu’s higher value reflects its load concentration: two buses (Honolulu and Ewa-Central) carry 72.4 % of the island. Concentrated load diversifies less.

3.2 A common-shock mixture

Independence across nodes is the strongest assumption in Model A and is physically implausible: one island shares one weather system and one clock. We therefore consider a mixture that couples the nodes.

Definition 3.3 (Model B). For a mixing weight $w \in [0, 1]$,

$$D_i = \mu_i(wE_0 + (1 - w)E_i),$$

with $E_0, E_1, \dots \stackrel{\text{iid}}{\sim} \text{Exp}(1)$; E_0 is a single shock shared by all nodes.

Proposition 3.4 (Induced moments). *Under Model B, for all $i \neq j$ in $\mathcal{B}^+$:*

$$\mathbb{E}[D_i] = \mu_i, \tag{6}$$

$$\text{CV}(D_i) = \sqrt{w^2 + (1 - w)^2} =: \kappa(w), \tag{7}$$

$$\text{Corr}(D_i, D_j) = \frac{w^2}{w^2 + (1 - w)^2} = \frac{w^2}{\kappa(w)^2}. \tag{8}$$

Furthermore κ is strictly convex on $[0, 1]$, symmetric about $w = 1/2$, and satisfies $\min_w \kappa(w) = \kappa(1/2) = 1/\sqrt{2}$ and $\kappa(0) = \kappa(1) = 1$.

Proof. (6) follows from $\mathbb{E}[E_0] = \mathbb{E}[E_i] = 1$. By independence and $\text{Var}(E) = 1$, $\text{Var}(D_i) = \mu_i^2(w^2 + (1 - w)^2)$, giving (7). For $i \neq j$ only the common term contributes to the covariance, $\text{Cov}(D_i, D_j) = \mu_i \mu_j w^2 \text{Var}(E_0) = \mu_i \mu_j w^2$, and dividing by $\mu_i \mu_j \kappa(w)^2$ gives (8). Finally $\kappa(w)^2 = 2w^2 - 2w + 1$ is a strictly convex parabola with vertex at $w = 1/2$ and value $1/2$ there, and value 1 at both endpoints. $\square$

Remark 3.5 (The mixing weight is not a correlation). Proposition 3.4 corrects a natural misreading. The parameter w is a mixing weight, not a correlation coefficient: at $w = 0.7$ the induced pairwise correlation is 0.845, not 0.7. More importantly, moving w away from 0 changes *two*

quantities at once—it raises correlation and, by the convexity in Proposition 3.4, *lowers* nodal dispersion from $\text{CV} = 1$ to $\kappa(0.7) = 0.762$. Model B is therefore a joint sensitivity, not a clean isolation of correlation. The construction also cannot produce nodal CV below $1/\sqrt{2} \approx 0.707$ at any w , so it cannot approach the empirical dispersion reported in Section 6. Simulation with 200,000 draws matches (7)–(8) to three decimal places at every $w \in \{0, 0.3, 0.5, 0.7, 1\}$.

3.3 Empirical resampling

Definition 3.6 (Model C). Let $h_1, \dots, h_H$ be the recorded hourly system totals ($H = 8760$) and let $s \in \Delta^{n_b}$ be a fixed allocation vector with $\sum_i s_i = 1$. Draw $K \sim \text{Unif}\{1, \dots, H\}$ and set $D = h_K s$.

Model C imposes no distributional assumption on the magnitude. It does, however, impose a strong assumption on the *direction*: $D/\|D\|_1 = s$ almost surely. Section 4 shows this is not innocuous, and Section 10 treats it as a limitation.

4 The geometry of failure

We now decompose a demand vector into magnitude and direction,

$$T = \|D\|_1 = \sum_i D_i, \quad \sigma = D/T \in \Delta^{n_b},$$

and ask how each of the two failure modes depends on the pair (T, σ) .

4.1 Adequacy is angular

Proposition 4.1 (Rays meet the servable set in an interval). *Fix $\sigma \in \Delta^{n_b}$ and define $T^*(\sigma) = \sup\{T \geq 0 : T\sigma \in \mathcal{S}\}$. Then $T^*(\sigma) < \infty$, the supremum is attained, and*

$$\{T \geq 0 : T\sigma \in \mathcal{S}\} = [0, T^*(\sigma)].$$

Proof. Finiteness. Summing (1e) over $i \in \mathcal{B}$, the flow terms cancel in pairs, giving $\sum_g P_g = \sum_i D_i = T$. With (1b) this forces $T \leq \sum_g p_g^{\max}$, so $T^*(\sigma) \leq \sum_g p_g^{\max} < \infty$.

Downward closure. Suppose $T\sigma \in \mathcal{S}$, witnessed by (P, θ, F) satisfying (1b)–(1f) with $S = 0$. Let $0 \leq T' \leq T$ and put $\lambda = T'/T \in [0, 1]$. We claim $(\lambda P, \lambda \theta, \lambda F)$ witnesses $T'\sigma \in \mathcal{S}$. Constraints (1c), (1d), (1e) are homogeneous of degree one in (P, θ, F, D) and therefore preserved under simultaneous scaling by λ , noting that the demand scales to $\lambda T\sigma = T'\sigma$ as required. For (1b), $0 \leq \lambda P_g \leq P_g \leq p_g^{\max}$ since $\lambda \in [0, 1]$ and $P_g \geq 0$. For (1f), $|\lambda F_\ell| \leq |F_\ell| \leq \bar{F}_\ell$. Hence $[0, T] \subseteq \{T' : T'\sigma \in \mathcal{S}\}$.

Attainment. $\mathcal{S}$ is the projection onto D -space of a polyhedron, hence a closed polyhedron; its intersection with the closed ray $\{T\sigma : T \geq 0\}$ is a closed bounded interval containing 0. $\square$

Corollary 4.2 (Exact characterisation of adequacy failure). *Load shedding is required at realisation $D \neq 0$ if and only if*

$$\|D\|_1 > T^*\left(\frac{D}{\|D\|_1}\right). \quad (9)$$

Consequently, for any demand model, $\mathbb{P}(\text{shed}) = \mathbb{P}(T > T^(\sigma))$, a comparison between a scalar magnitude and a direction-dependent threshold.*

Corollary 4.2 makes precise the sense in which adequacy is angular: the bar that magnitude must clear is itself a function of direction. A demand model that holds direction fixed—Model C does so by construction—collapses T^* to a constant and turns adequacy into a pure magnitude threshold.

Table 2: Nameplate versus deliverable capacity along the design direction $\bar{\sigma}$. The final column is capacity that exists but cannot reach load.

System	Nameplate (MW)	Deliverable (MW)	Stranded (MW)	Stranded (%)
IEEE 14-bus	772.4	575.1	197.3	25.5
O‘ahu 9-bus	1508.1	1082.0	426.1	28.3

Definition 4.3 (Deliverable capacity). Write $\bar{\sigma} = \mu/M$ for the design direction. The *deliverable capacity* of the network is $T^*(\bar{\sigma})$, computed by the linear program

$$T^*(\bar{\sigma}) = \max \left\{ T : (1b)-(1f) \text{ hold with } D = T\bar{\sigma}, S = 0 \right\}, \quad (10)$$

which is well posed because T enters (1e) linearly.

Deliverable capacity should be contrasted with nameplate capacity $\sum_g p_g^{\max}$. The proof of Proposition 4.1 shows $T^*(\sigma) \leq \sum_g p_g^{\max}$ always; the gap is generation that exists but cannot be moved to where demand sits. Table 2 reports both systems.

Remark 4.4 (An artefact worth naming). On IEEE-14 the stranded 197.3 MW has an identifiable cause. Bus 8 carries a generator and no load, and is radially connected. In the base case its branch therefore carries zero flow, so the rating heuristic of Section 6—size each line at $1.5\times$ its base-case duty, floored at 25 MW—assigns it the floor. Seventy-five of that unit’s 100 MW can never leave the bus. This is not an error so much as an honest consequence of sizing wires to present duty, but it means a quarter of the test system’s headline capacity is unreachable by construction.

4.2 Congestion is radial

Write $\mathcal{C}$ for the set of demand vectors at which at least one branch attains its thermal limit in the optimal dispatch. Our initial conjecture was that $\mathcal{C}$, like $\mathcal{S}^c$, is governed primarily by direction. The following conditional experiment refutes it.

Proposition 4.5 (Congestion is approximately a magnitude threshold). *On the O‘ahu network there exists t_c such that, for the two demand models A and C,*

$$\mathbb{P}(D \in \mathcal{C}) \approx \mathbb{P}(\|D\|_1 > t_c),$$

uniformly in the model. Numerically $t_c = 585$ MW, and the approximation is accurate to 0.6 percentage points for both models despite their congestion frequencies differing by 34 points.

This is an empirical statement about a particular network, established by computation rather than proof, and we state it as such. The evidence is in Table 3. Conditioning on total demand removes essentially all of the difference between the two models: in every band above 600 MW both models congest with probability at or near one. The unconditional gap (58.40% versus 92.51%) is therefore almost entirely attributable to *where each distribution places its mass* relative to a fixed threshold—the exponential spends 42.8% of its draws below 600 MW, the empirical record only 11.3%.

The right-hand block of Table 3 shows the contrast that motivates Corollary 4.2. At *identical* total demand in the 900–1000 MW band, Model A sheds in 20.9% of draws while Model C never sheds at all. Magnitude does not determine adequacy; direction does.

Table 3: Failure probabilities conditional on total demand, O‘ahu, 10,000 draws per model. Congestion is insensitive to the demand model once magnitude is controlled; shedding is not.

Band T (MW)	$\mathbb{P}(\text{congestion} \mid T)$		$\mathbb{P}(\text{shedding} \mid T)$	
	Model A	Model C	Model A	Model C
600–700	93.4%	100.0%	1.9%	0.0%
700–800	99.7%	100.0%	2.8%	0.0%
800–900	100.0%	100.0%	6.6%	0.0%
900–1000	100.0%	100.0%	20.9%	0.0%
1000–1100	—	—	45.0%	0.0%
<i>Radial decomposition with $t_c = 585$ MW:</i>				
Model A	$\mathbb{P}(T > t_c) = 59.0\%$		observed 58.4%	(gap 0.6 pp)
Model C	$\mathbb{P}(T > t_c) = 92.0\%$		observed 92.5%	(gap -0.5 pp)

Table 4: Distribution of the directional threshold $T^*(\sigma)$ on the O‘ahu network, 1200 directions drawn from Model A, compared with the fixed design direction $\bar{\sigma}$ used by Model C.

Quantity	T^* (MW)
1st percentile	591.7
5th percentile	740.1
25th percentile	927.2
Median	1027.7
75th percentile	1108.8
95th percentile	1168.8
Mean	1005.8
Minimum	409.1
Maximum	1496.1
Fixed design direction $T^*(\bar{\sigma})$	1082.0
Observed annual peak $\max_k h_k$	1054.3

4.3 The distribution of the directional threshold

Corollary 4.2 invites a direct question: how variable is $T^*(\sigma)$? We computed (10) for 1200 directions drawn from Model A; Table 4 reports the resulting distribution alongside the fixed-direction value.

Two readings matter. First, T^* is genuinely variable: its interquartile range spans 182 MW, and its 1st percentile lies below 600 MW, meaning that roughly one direction in a hundred cannot deliver even the island’s *average* load. Second, the design direction is comparatively favourable: $\mathbb{P}(T^*(\sigma) < T^*(\bar{\sigma})) = 64.7\%$, so a randomly oriented demand vector is more often harder to serve than the population-share direction. This is the mechanism behind the zero shedding observed under Model C, and it also cautions that the zero is partly a property of the allocation assumption.

Theorem 4.6 (Why Model C never sheds). *Under Model C, $\mathbb{P}(\text{shed}) = 0$ if and only if $\max_{k \leq H} h_k \leq T^*(s)$. On the O‘ahu data, $\max_k h_k = 1054.3$ MW and $T^*(\bar{\sigma}) = 1082.0$ MW, so the condition holds with a margin of 27.7 MW, or 2.63% of peak.*

Proof. Under Model C every realisation lies on the ray $\{Ts : T \geq 0\}$ with $T = h_K$. By Corollary 4.2, shedding is required at realisation k iff $h_k > T^*(s)$. Hence no realisation sheds iff $\max_k h_k \leq T^*(s)$. The numerical values follow from (10) and the load record of Section 6. $\square$

Theorem 4.6 converts an empirical zero into a structural statement with a margin attached, and that margin is the number a planner should care about. A nameplate reserve margin of $(1508.1 - 1054.3)/1054.3 = 43.0\%$ is comfortable. The deliverable margin of 2.63% is not.

5 Monte Carlo design

5.1 Estimators and uncertainty

For an event A (shedding, congestion, dispatch of the costliest tier) we estimate $p = \mathbb{P}(D \in A)$ by $\hat{p}_N = N^{-1} \sum_{k=1}^N \mathbf{1}\{D^{(k)} \in A\}$ with $D^{(k)}$ iid from the demand model. Since Proposition 2.5 guarantees every replication returns a well-defined answer, $\hat{p}_N$ is unbiased with $\text{Var}(\hat{p}_N) = p(1-p)/N$. We report Wilson score intervals, which behave far better than the Wald interval near $p = 0$ —relevant here, since Model C produces $\hat{p} = 0$ exactly:

$$\frac{\hat{p} + \frac{z^2}{2N} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{N} + \frac{z^2}{4N^2}}}{1 + \frac{z^2}{N}}. \quad (11)$$

At $N = 10,000$ and $z = 1.96$ the half-width is at most 0.98%, and at $\hat{p} = 0$ the interval is $[0, 3.84 \times 10^{-4}]$.

5.2 Degeneracy and tie-breaking

The IEEE-14 cost data assigns two generators the same marginal cost. When units tie, the optimal face of the linear program is not a singleton: the *total* output of the tied pair is determined but the split between them is not, and which vertex a solver reports is decided by its pivoting rule. Probing the optimal face confirmed that one generator's output can move by up to 140 MW at unchanged objective value. A per-generator statistic computed naively would therefore report solver internals as if they were model output.

Proposition 5.1 (Invariance under tie-breaking). *Replace c_g by $c_g + \varepsilon_g$ with $0 < \varepsilon_g < \delta$, where δ is smaller than the least nonzero gap between distinct attainable objective values. Then the perturbed program has a unique optimal P^* for generic ε , and the quantities $\sum_i S_i^*$, $\mathbf{1}\{\text{congestion}\}$ and $\sum_{g \in \mathcal{T}} P_g^*$ —where $\mathcal{T}$ is any set of generators sharing a common cost—are unchanged from the unperturbed program.*

Proof sketch. The perturbation changes the objective by at most $\delta \sum_g p_g^{\max}$, which by the choice of δ is insufficient to make a previously suboptimal vertex optimal; hence the perturbed optimum lies on the original optimal face. On that face the listed quantities are constant, because each is determined by the aggregate balance (1e) and the flow variables, which are common to all points of the face when the cost coefficients within $\mathcal{T}$ coincide. Uniqueness for generic ε is the standard lexicographic argument. $\square$

We use $\varepsilon_g = 10^{-6}g$, which perturbs the objective by well under one thousandth of a dollar against hourly costs in the thousands. Consistent with Proposition 5.1, the headline probabilities move by at most 0.5 percentage points under this change, and the aggregate cost and expensive-tier output are unchanged.

5.3 Computation

The structural data of $\mathcal{P}(D)$ —topology, susceptances, ratings, generator bounds, objective coefficients—is identical across replications; only the right-hand side of (1e) and the upper

Table 5: The O‘ahu load record, 8760 filed hourly observations.

Statistic	Value (MW)
Mean net load	743.86
Median	720.75
Standard deviation	124.91
Minimum	472.58
Maximum (annual peak)	1054.26
Coefficient of variation	0.1678
Load factor (mean/peak)	0.706
Installed nameplate capacity	1508.1 MW

bound in (1g) vary. We therefore build the program once and re-parameterise it, which permits the solver to warm-start from the previous basis. Ten thousand replications complete in approximately five seconds. Occasional numerical failures under sustained warm-starting are handled by rebuilding the model from a cold basis and re-solving; this was triggered on fewer than one replication in a thousand and never changed a reported quantity.

6 Systems and data

6.1 The O‘ahu model

The O‘ahu system is assembled from primary filings. Hourly load is reconstructed from Hawaiian Electric’s Final O‘ahu Inputs Workbook 3 (filed 2022), 2021 base scenario, giving $H = 8760$ observations of system net load. The dispatchable fleet ($n_g = 24$) is taken from EIA Form 860 summer capacities with implied heat rates from EIA Form 923 and fuel prices from the utility’s own forecast. The network is a stylised nine-bus representation of the published 138 kV two-corridor ring, with buses at the seven Census county subdivisions (the Ewa subdivision split to expose generation siting) and loads allocated by resident population. Appendix A gives full provenance.

The single most consequential number in Table 5 is the coefficient of variation. The empirical record has $CV = 0.168$; the Exp(1) marginals of Model A impose nodal $CV = 1$. Figure 1 displays the two distributions on a common mean.

Figure 2 shows the fleet’s merit order. Because real demand is confined to a narrow band, it terminates in a narrow range of the cost stack: an average hour is served to $126.40 \text{ MW}^{-1} \text{ h}$ and the annual peak only to $134.48 \text{ MW}^{-1} \text{ h}$, leaving the diesel peakers at $222 \text{ MW}^{-1} \text{ h}$ to $238 \text{ MW}^{-1} \text{ h}$ untouched.

6.2 The IEEE 14-bus system and its calibration

The IEEE 14-bus case is used as a controlled comparison. Two adaptations are required. First, the published case lists no thermal ratings; since an unconstrained network can never congest, ratings are assigned as $\bar{F}_\ell = \max\{25, \lceil 1.5 |F_\ell^{\text{base}}| \rceil\}$ from a base-case solve. Second, and more importantly, the two systems must be placed under comparable stress.

Definition 6.1 (Utilisation). $\rho = M / \sum_g p_g^{\text{max}}$, the ratio of design mean demand to nameplate capacity.

All three questions Q1–Q3 concern headroom, and headroom is governed by ρ . Comparing a test system at its native $\rho = 0.335$ against an island at $\rho = 0.493$ would compare calibrations rather than networks. We therefore set the IEEE-14 mean to $\rho_{\text{O‘ahu}} \times 772.4 \text{ MW} = 380.98 \text{ MW}$,

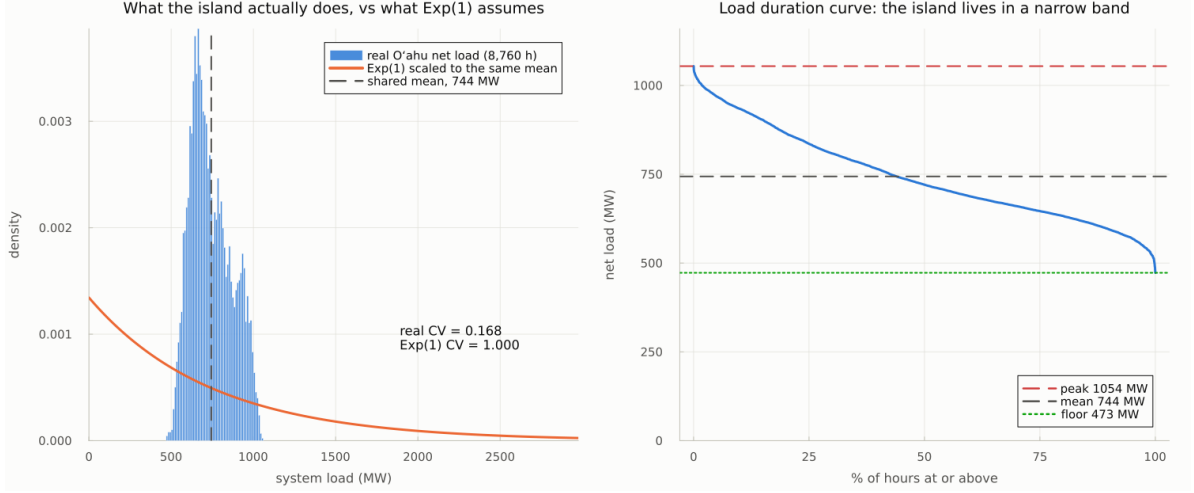


Figure 1: Left: the empirical distribution of O’ahu net load against an exponential density scaled to the same mean of 743.9 MW; the comparison is therefore purely one of shape. Right: the load duration curve. The record occupies a band of roughly 0.64 to 1.42 times its own mean and never leaves it.

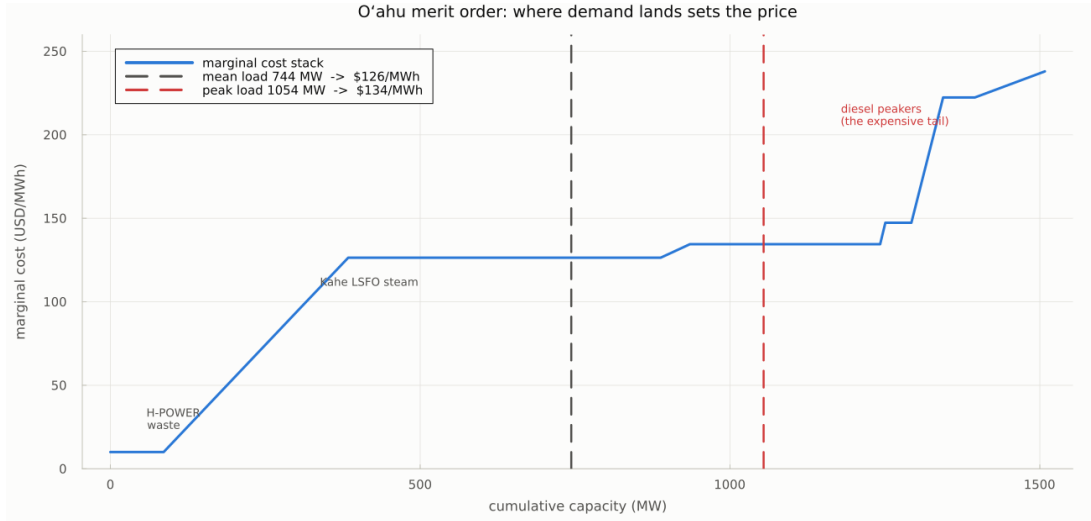


Figure 2: O’ahu merit order. Cumulative capacity on the abscissa, marginal cost on the ordinate. The step on which demand terminates is the system marginal price.

distributed across the eleven load buses in the published proportions. The native case is retained as a reference point.

Remark 6.2 (The linear cost simplification). By Assumption 2.2(iii) we use only the linear cost coefficient, discarding the quadratic term present in the IEEE-14 data. This keeps the program a linear program with genuine locational marginal prices as duals, and matches the convention used throughout the wider project. It has one consequence for Q1: under linear costs the three units at buses 3, 6 and 8 share the maximum cost of $40 \text{ MW}^{-1} \text{ h}$, so “the most expensive generator” is properly a most expensive *tier*. Under the full quadratic cost the incremental costs at full output would rank the units differently. We report the tier and label it as such.

7 Results: the IEEE 14-bus system

Table 6 reports the three headline probabilities at $N = 10,000$, with Wilson intervals.

The most informative comparison in Table 6 is between shedding frequency and the fre-

Table 6: IEEE 14-bus under Model A at $\rho = 0.493$, $N = 10,000$, seed 20,260,908. Intervals are Wilson score at 95%.

Event	Estimate	95% interval
Costliest tier dispatched	59.63%	[58.66, 60.59]
Load shedding required	41.58%	[40.62, 42.55]
At least one branch at its limit	60.95%	[59.98, 61.91]
Mean branches at limit per replication	1.39 of 20	
Energy not served, share of demand drawn	8.29%	
Realised system CV	0.444	

Table 7: O’ahu, $N = 10,000$ per model, network identical throughout.

Demand model	System CV	Costliest unit	Shedding	$\mathbb{E}[\text{shed}]$	Branch at limit
A independent Exp(1)	0.532	5.48%	19.77%	56.8 MW	58.40%
B common shock, $w = 0.7$	0.708	5.20%	19.53%	85.4 MW	51.05%
C empirical resampling	0.168	0.00%	0.00%	0.0 MW	92.51%

quency with which demand exceeds capacity. Demand exceeds *nameplate* capacity in only 2.87 % of replications, and exceeds *deliverable* capacity in 12.1 %, yet shedding occurs in 41.58 %. Most adequacy failures on this network are therefore failures of delivery, not of generation—consistent with Corollary 4.2, since a randomly oriented demand vector frequently has $T^*(\sigma)$ far below $T^*(\bar{\sigma})$.

Sensitivity to utilisation. The lower-right panel of Figure 3 sweeps ρ from 0.25 to 0.79 with the network—crucially including its thermal ratings—held fixed. Shedding probability rises from 3.95 % to 79.30 % across that range. An earlier version of this sweep re-derived ratings at each ρ from the corresponding base case, which silently enlarged the network as demand grew and produced a curve roughly four times too flat at the low end. We record the error because it is easy to commit: a sweep over demand must not re-tune the network.

Sensitivity to the random stream. Repeating the full experiment on four independent seeds gives shedding probabilities 41.58, 42.32, 42.73 and 43.10 %—a spread of 1.52 points against a Wilson half-width of 0.97, consistent with binomial sampling error. The reported seed, fixed in advance as the presentation date, sits at the low end of this range, so the headline figures are mildly conservative.

8 Results: the O’ahu system

Table 7 reports all three demand models on the unchanged O’ahu network.

Model B is instructive precisely because it moves so little. Introducing a common island-wide shock leaves shedding *frequency* essentially unchanged (19.53 against 19.77 %) while raising expected unserved energy by half (85.4 against 56.8 MW). Correlation, on this network, is a statement about the depth of failures rather than their incidence. As Remark 3.5 notes, the comparison is not a clean isolation of correlation, since w also lowers nodal dispersion.

The binding corridor. Under Model C the branch from Ewa-West to Ewa-Central operates at a mean 97.7 % of its 400 MW rating and is at its limit in 68.8 % of replications. Its structural role is visible from the topology: the Kahe (582 MW) and Ewa-West (419 MW) generation

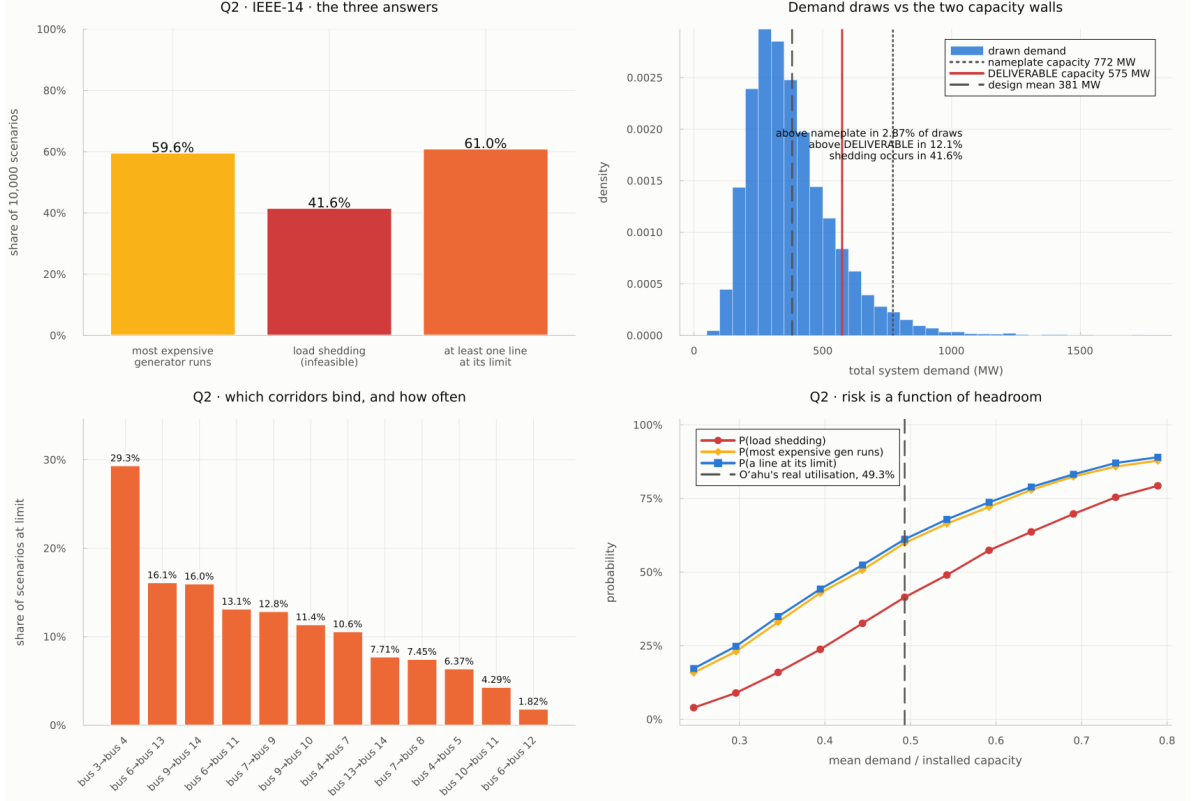


Figure 3: IEEE 14-bus results. Top right: drawn demand against both capacity walls. Bottom left: branch-level congestion frequency; the corridor from bus 3 to bus 4 binds in 29.3% of replications. Bottom right: the three probabilities as functions of utilisation, with the network held fixed.

sits behind it, and 84% of island load sits in front of it. This single branch is what reduces deliverable capacity to 1082.0 MW and, by Theorem 4.6, what sets the 2.63% adequacy margin.

9 Misspecification: relocation rather than inflation

The comparison between rows A and C of Table 7 is the paper’s empirical centre. Holding the network, the solver, the estimator and the sample size fixed, and changing only the demand distribution:

Quantity	Assumed Exp(1)	Empirical	Direction
$\mathbb{P}(\text{shedding})$	19.77%	0.00%	↓ to zero
$\mathbb{P}(\text{costliest unit})$	5.48%	0.00%	↓ to zero
$\mathbb{P}(\text{branch at limit})$	58.40%	92.51%	↑ by 34 points

The two failure modes move in *opposite directions*. This is the sense in which the misspecification relocates risk rather than inflating it, and Section 4 supplies the mechanism for each half.

Why adequacy risk vanishes. By Corollary 4.2 shedding requires $T > T^*(\sigma)$. Model A draws directions with $T^*(\sigma)$ frequently well below the design value (Table 4: median 1027.7 MW, first percentile 591.7 MW) and simultaneously draws magnitudes with a heavy right tail, so the two cross often. Model C fixes the direction at a comparatively favourable $\bar{\sigma}$ and caps the magnitude at the observed peak; by Theorem 4.6 the crossing never occurs.

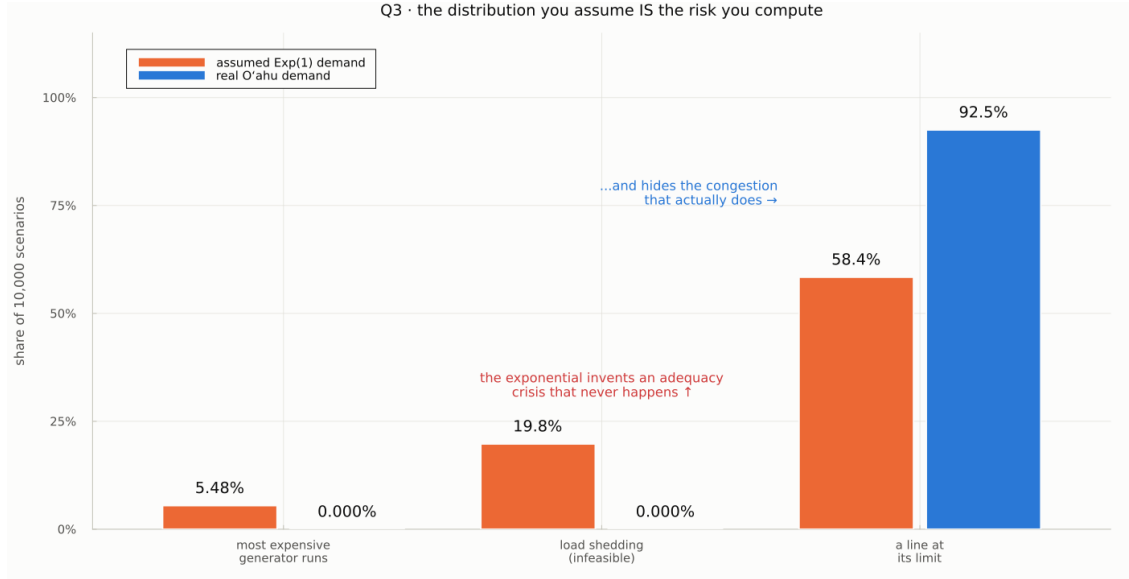


Figure 4: The three questions answered under the assumed exponential demand and under the empirical record, on an identical network.

Why congestion risk rises. By Proposition 4.5 congestion is approximately the event $T > t_c$ with $t_c = 585$ MW. The exponential places 42.8% of its mass below that threshold; the empirical record, which never falls below 472.6 MW and has a load factor of 0.706, places only 11.3% there. Real load sits persistently in the band that saturates the network, whereas the exponential repeatedly wanders out of it.

A general caution. The direction of misspecification bias is therefore mode-dependent, and cannot be signed a priori from the tail weight of the assumed distribution. A heavier tail increases the frequency of events defined by an upper threshold on a *radial* quantity (here, adequacy failure combined with the magnitude crossing) but may either increase or decrease the frequency of events defined by an *interval* in that quantity. Congestion is of the second type: the network is uncongested both when demand is small and—because an adequacy failure is resolved by shedding at the constrained bus—the relationship at the top of the range is not monotone either. Practitioners sometimes justify fat-tailed demand models as “conservative”. On the evidence here that justification holds for adequacy and fails for congestion.

10 Limitations

We list the limitations that bear directly on the conclusions.

- L1. Model C fixes the demand direction.** By construction $D = h_K s$, so all directional variation is suppressed. The zero shedding of Theorem 4.6 is thus a joint statement about the magnitude record *and* the population-share allocation, and Table 4 shows the allocation is comparatively favourable. A resampling scheme preserving the empirical spatial copula would be the correct next step, and would likely produce a small but nonzero shedding probability. The 2.63% margin is the honest headline, not the zero.
- L2. Bus-level loads are proxied by population.** The allocation s is resident population share. Honolulu’s commercial, hotel and military load is not proportional to residence, so the true direction almost certainly differs from $\bar{\sigma}$, and could be less favourable.

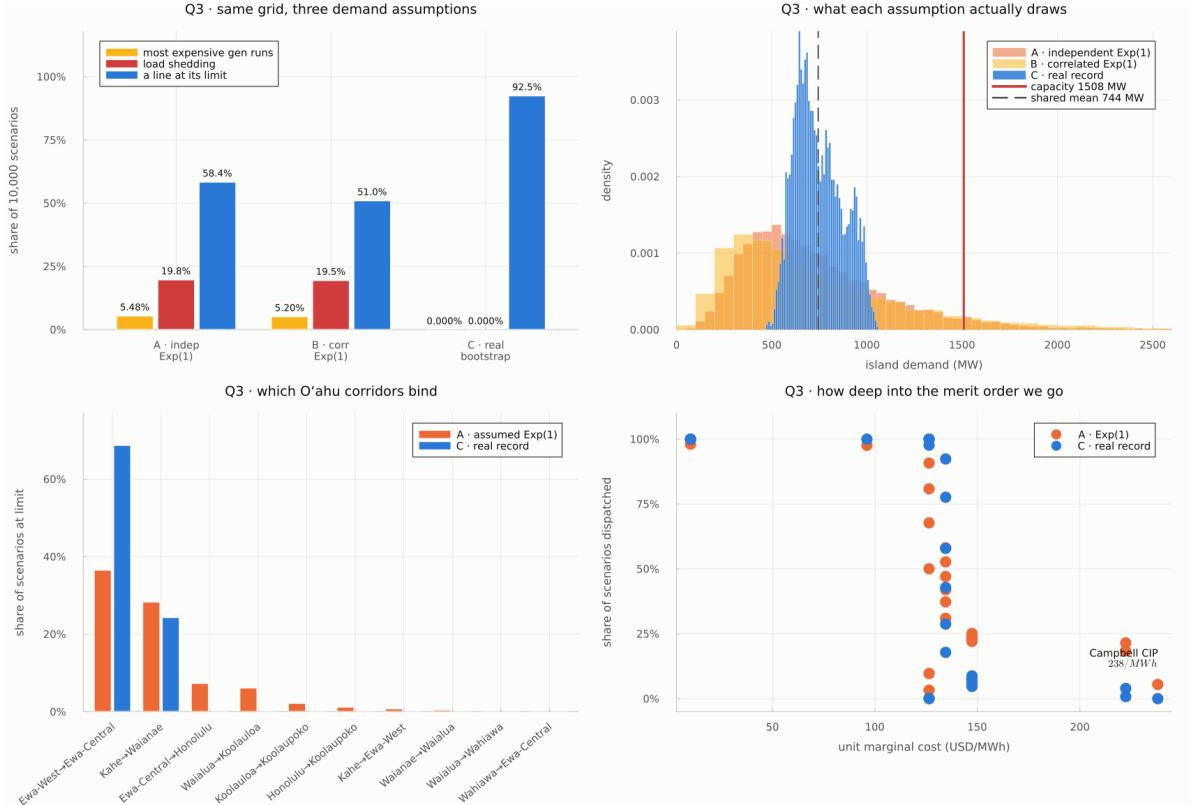


Figure 5: O’ahu under all three demand models: failure frequencies, the demand distributions each model actually draws, branch-level congestion, and depth of penetration into the merit order.

- L3. The network is stylised.** Nine buses on a two-corridor 138 kV ring with 200 MW per circuit is a labelled technology assumption, not the utility’s model. The *ranking* of corridors is a robust finding; the exact percentages inherit the assumption. The Texas A&M HAWAII40 synthetic case is the natural cross-check.
- L4. No unit commitment.** Assumption 2.2(ii) sets $p^{\min} = 0$. Real minimum stable output, start-up costs and minimum up/down times would raise costs and could create adequacy failures at low demand that this model cannot see.
- L5. DC approximation and no contingencies.** Losses, reactive power and voltage limits are absent, and the study is $N=0$. Given that the central finding is a single binding corridor, an $N=1$ analysis is the most valuable extension: the deliverable capacity of Table 2 would fall further under credible outages.
- L6. Congestion means saturation, not violation.** In a hard-constrained formulation flows cannot exceed ratings; the constraint binds instead. Our congestion statistic measures saturation. A companion unconstrained solve measures how far over a rating flow would have gone, and is reported in the repository.

11 Conclusion

Three conclusions survive the analysis.

First, the recourse formulation is the right instrument. Pricing unserved energy at a value of lost load that exceeds the maximum nodal dual—not merely the maximum generator cost, per Remark 2.7—makes the program feasible for every realisation and converts adequacy failure into a measurable magnitude.

Second, the two failure modes have different geometry. Adequacy failure is exactly the event $\|D\|_1 > T^*(D/\|D\|_1)$ (Corollary 4.2), a comparison against a direction-dependent threshold; congestion is, on this network, approximately a fixed threshold on magnitude alone (Proposition 4.5). A demand model is therefore not adequate or inadequate in general—it is adequate or inadequate *for a given failure mode*.

Third, the practical finding. Replacing an assumed $\text{Exp}(1)$ demand with 8760 hours of filed data eliminates an apparent 19.77 % shedding probability while raising congestion frequency by 34 percentage points. And the reserve margin that matters is not the 43.0 % computed against nameplate capacity but the 2.63 % computed against what the network can actually deliver—a margin set by one 400 MW corridor, behind which 426 MW of generation sits stranded.

Acknowledgements. This work was carried out as part of an ongoing research collaboration with Adrian DeClerq, whose framing of the original question shaped the study.

A Data provenance

Table 8: Primary sources for the O‘ahu model.

Model element	Source
Hourly load, 8760 obs.	Hawaiian Electric, Final O‘ahu Inputs Workbook 3, filed 2022-03-31; 2021 base scenario, reconstructed from underlying load, managed EV, distributed PV, storage, efficiency and time-of-use layers.
Unit list, capacities	U.S. EIA Form 860 (2024 final), summer capacity; operational Honolulu County utility and contracted firm resources.
Heat rates	U.S. EIA Form 923 (2024 final): electric fuel consumption divided by net generation, at plant/prime-mover level.
Fuel prices	Hawaiian Electric 2024 base forecast (LSFO, diesel, ULSD).
Buses and load allocation	U.S. Census 2020 P.L. 94-171 county subdivisions via TIGERweb; population-weighted tract centroids; loads allocated by resident population.
Topology	Hawaiian Electric published <i>Power Delivery</i> description: two-corridor 138 kV ring; nine buses, ten branches, two independent loops.
Electrical parameters	Technology defaults at 138 kV, labelled as assumptions in the data files: $0.8\ \Omega/\text{mile}$ reactance, 200 MW per circuit.
IEEE 14-bus	MATPOWER <code>case14</code> ; ratings assigned per Section 6.

B Reproducibility

All results are produced by a Julia implementation using JuMP and the HiGHS simplex solver. The random seed is 20,260,908 throughout. The computational sequence is:

```
julia "Q1 - hawaii data.jl"          # data audit, Figures 1-2
julia "Q2 - ieee14 montecarlo.jl"    # Table 5, Figure 3
julia "Q3 - oahu montecarlo.jl"      # Table 6, Figures 4-5
```

julia anim_set8.jl

network animation

The engine (`montecarlo.jl`) implements Definition 2.3, the three demand models of Section 3, the deliverable-capacity program (10), and the Wilson interval (11). Case construction for both networks is in `cases.jl`. Every figure and table in this paper regenerates from those scripts; intermediate results are written to `results/` as CSV.

Verification. The formulation was checked against an independent re-implementation of the power flow written from first principles. Over 900 solved replications, energy balance closed to 6×10^{-12} MW, nodal Kirchhoff residuals recomputed from scratch to 1.2×10^{-11} MW, and the worst thermal-limit violation was 1.9×10^{-12} MW. The moment identities of Propositions 3.2 and 3.4 were confirmed against 200,000-draw simulations; the interval structure of Proposition 4.1 was confirmed by fine scans along both design rays with no violations; and Proposition 2.6 was checked on 6000 replications with no counterexamples.